

Career Overview

Enterprise AI Architect and SRE Leader with nearly 20 years of experience bridging the gap between cutting-edge AI research and production-grade, large-scale operations. Proven track record of architecting, operationalizing, and scaling mission-critical platforms for conversational intelligence, multimodal streaming, and low-latency inference. Specialized in building deterministic, high-availability AI systems that unify symbolic logic with neural frameworks to eliminate operational friction, automate decisioning, and drive enterprise transformation.

Skills:

Enterprise AI & Solution Architecture	Enterprise AI Architecture, AI Platform Engineering, Distributed Systems, Production AI, End-to-End SDLC, Productionization, Operationalization, Enterprise Integration, Scalable System Design, Reliability Engineering
Agentic AI & LLM Engineering	Agentic AI, Multi-Agent Systems, LangGraph, MCP, RAG, Prompt Engineering, Structured Outputs, Tool Calling, AI Orchestration, Conversational AI, Intent Classification, Slot Filling, FastMCP
Machine Learning & AI Engineering	PyTorch, TensorFlow, scikit-learn, MLflow, Model Serving, Model Lifecycle Management, MLOps, AI Governance, Explainable AI (XAI), Model Risk Management, Drift Detection, Confidence Scoring
AI/ML Techniques	Predictive Analytics, Statistical Modeling, Multimodal AI, Feature Engineering, Anomaly Detection, Neurosymbolic AI, Explainability, NL-to-Action, NL-to-Inference, Model Validation, Post-Inference Decisioning
Data Engineering & Pipelines	Apache NiFi, Apache Airflow, Hadoop, Spark, Streaming Pipelines, Batch Processing, Data Quality, Data Lineage, Data Governance, Compliance, ETL/ELT
Programming & APIs	Python, SQL, FastAPI, REST APIs, Microservices
Cloud & Platform Engineering	AWS, Azure, GCP, Docker, Kubernetes, OpenShift, Infrastructure as Code (Terraform, Ansible)
DevOps, CI/CD & Platform Automation	GitHub Actions, Jenkins, Concourse, Bamboo, CI/CD Pipelines, Release Automation, Deployment Automation, Infrastructure Automation
Observability & Reliability Engineering	Grafana, ELK, Splunk, AppDynamics, Telegraf, Prometheus, VictoriaMetrics, Consul, Alerta, Health Scoring, SLO/SLI, Alert Intelligence, Signal-to-Noise Optimization
Databases & Storage	PostgreSQL, MSSQL, Oracle, MySQL, MongoDB, Couchbase
Security & Governance	Application Security, Infrastructure Security, AI Governance, Compliance Frameworks, Access Control, OS & Database Hardening, CMDB
Leadership & Delivery	Technical Leadership, Enterprise Architecture, Engineering Strategy, Stakeholder Management, Program Delivery, Team Leadership, Vendor Management, Risk Management, Mentoring
Enterprise AI & Solution Architecture	Enterprise AI Architecture, AI Platform Engineering, Distributed Systems, Production AI, End-to-End SDLC, Productionization, Operationalization, Enterprise Integration, Scalable System Design, Reliability Engineering

Experience:**Enterprise AI & Platform Architecture**

- Owned end-to-end AI platform delivery spanning multi-turn conversational AI, agentic workflows, real-time ML inference, and intelligent infrastructure automation for high-throughput production environments.

- Led the design and deployment of cloud-native AI services using FastAPI, LangGraph, and Model Context Protocol (MCP), enabling the secure production deployment of real-time AI workflows across enterprise environments.
- Established enterprise AI architecture standards covering risk management, explainability, model lifecycle governance, Token Optimization and production readiness to ensure compliance across distributed engineering teams.
- Led enterprise AI modernization by integrating LLMs, multi-agent orchestration, and deterministic decisioning into production platforms that improved operational automation and governance.
- Partnered with business and engineering leaders to align technical roadmaps with corporate strategy, translating complex operational friction points into scalable, production-ready AI solutions.

AI Engineering & Machine Learning

- Delivered production-grade conversational AI systems supporting multi-turn dialogue management, intent classification, slot extraction, and tool calling to automate complex backend workflows.
- Developed real-time multimodal feature fusion architectures combining streaming text, audio, and behavioral signals, which optimized prediction accuracy and enhanced live decision quality.
- Designed explainable AI (XAI) frameworks incorporating confidence scoring and deterministic validation, directly improving trust and driving operational adoption in regulated environments.
- Built intelligent decisioning frameworks that reconciled statistical baselines with machine learning anomaly detection, reducing operational noise and eliminating false positives.
- Drove enterprise LLMOps and MLOps capabilities, implemented continuous experiment tracking, automated validation gates, and live drift detection to protect model integrity post-deployment.

Enterprise Platform Engineering

- Architected data integration pipelines to aggregate operational telemetry, CMDB datasets, and cloud streams into unified, AI-ready data layers.
- Developed reusable cloud-native backend microservices supporting scalable inference pipelines, API orchestration, and seamless enterprise system integration.
- Spearheaded enterprise-wide platform modernization across CI/CD infrastructure, Kubernetes, OpenShift, hybrid cloud footprints, and Infrastructure as Code (Terraform/Ansible).
- Managed over 300 production CI/CD pipelines supporting 6,000+ automated jobs, increasing engineering velocity and establishing delivery consistency across global teams.
- Built reusable engineering frameworks that standardized security hardening, automated OS patching, proactive monitoring, and compliance monitoring across legacy and next-gen platforms.

Reliability Engineering & Observability

- Owned the enterprise observability strategy covering application performance, infrastructure health, AI system monitoring, and real-time operational intelligence.
- Designed AI-driven SRE platforms combining ML anomaly detection, statistical thresholds, and automated health scoring to preserve 99.99% availability for mission-critical core applications.
- Directed production incident response, chaos engineering simulations, and operational readiness war games to proactively identify vulnerabilities and benchmark system resilience under stress.
- Optimized signal-to-noise ratios through intelligent alert prioritization, automated scoring, and signal correlation frameworks, eliminating alert fatigue for engineering teams.
- Instituted automated quality gates and capacity planning guidelines, partnering with distributed engineering groups to enforce site reliability engineering (SRE) best practices.

Leadership & Enterprise Delivery

- Provided cross-functional technical leadership and architectural accountability across high-impact enterprise AI, cloud infrastructure, DevOps, and site reliability engineering initiatives.

- Led globally distributed, multi-timezone technology teams of 50+ members, managing hiring, onboarding, mentoring, and technical excellence across full project lifecycles.
- Influenced senior executive stakeholders on technology strategy, production AI operating models, vendor management, and enterprise platform modernization roadmaps.
- Established corporate engineering standards covering software quality, system security, budget/cost optimization, and delivery risk governance.
- Cultivated a high-velocity culture focused on automation, continuous improvement, and mentorship, driving technical growth for engineers, architects, and technical leads.

Education:

- Master of Science (MS), Artificial Intelligence & Machine Learning – In Progress
 - Drexel University, Philadelphia, PA — Expected 2028
- Bachelor of Engineering (BE), Electrical & Electronics Engineering – 2006

Professional Experience:

- Comcast Cable Communications Management LLC
 - Eng 6, Software Development & Engineering | Dec 2019 – Present
- Cognizant Technology Solutions
 - Delivery Manager | Jun 2013 – Dec 2019
- Huawei Technologies India Pvt. Ltd.
 - Technical Project Leader | Feb 2009 – May 2013
- Keane International Pvt. Ltd.
 - Senior Software Engineer – Quality | Jul 2006 – Jan 2009